Week 22: Meeting regions

Adult guide built from the problem files · Unpiloted. Physical pretests unperformed.

This week's adult guide is week-22-facilitator.pdf, whose source is editable/facilitator-src/build_guide.py. These problem files hold each problem's one-sentence idea, not its answers; the answers, hints and proofs are in the guide.

Problems and routes

Problem	K–1	2–3	4–5	route
Put D on each cross and make the regions meet	P1	P1	P1	first
Every meeting split of four dots on a line			P2	first
Place D against a fixed triangle	P2			first
The placing game: can four dots always be split?	P3	P2	P3	first
Two meeting groupings for three places of D	P4			first
Which crosses for C make two regions meet?	P5			return
Meeting splits of four dots on a line	P6	P3		K-1 return; 2-3 first
A shared segment, but never a shared filled triangle		P4		first
Two arrangements with no common working split		P5		first
Three dots with no meeting split		P6		return
Which crosses for C give a meeting split?		P7		return
Two labels at one point			P4	return
Arrangements with one, and more than one, working split			P5	return
Can every arrangement be split?			P6	return
Three dots with no meeting split			P7	return
The smallest number of dots that guarantees a split			P8	return

First route. K–1 P1–P4; Grades 2–3 P1–P5; Grades 4–5 P1–P3. **Return visits.** K–1 P5–P6; Grades 2–3 P6–P7; Grades 4–5 P4–P8.

Key, one entry per problem

Put D on each cross and make the regions meet

K–1 P1 · Grades 2–3 P1 · Grades 4–5 P1 · first route · wording differs by band

Idea. For every position of D some split works (Radon): D inside triangle ABC gives D against A, B, C; otherwise two segments cross or a point lies on a segment.

Every meeting split of four dots on a line

Grades 4–5 P2 · first route

Idea. For four dots on a line, exactly four of the seven splits work: either middle dot alone, and the two pair splits that interleave or nest; an end dot alone, or the left pair against the right pair, fails.

Place D against a fixed triangle

K–1 P2 · first route

Idea. Wherever D goes, the other player can make regions meet: D inside the triangle stands alone, and outside it two segments cross or D lies on a side line.

The placing game: can four dots always be split?

K–1 P3 · Grades 2–3 P2 · Grades 4–5 P3 · first route · wording differs by band

Idea. No arrangement of four dots defeats the splitter (Radon's theorem in the plane): one dot in the triangle of the others, two crossing segments, or a degenerate case always gives a meeting split.

Two meeting groupings for three places of D

K–1 P4 · first route

Idea. With D on segment AB, beyond B on that line, or between A and C, exactly two splits work each time (AB | CD and ABC | D; AD | BC and ACD | B; AC | BD and ABC | D).

Which crosses for C make two regions meet?

K–1 P5 · return route

Idea. With three dots, regions meet only when C lies on the line AB (between, beyond, or on A or B); anywhere off that line no split works.

Meeting splits of four dots on a line

K–1 P6 · Grades 2–3 P3 · K-1 return; 2-3 first route · wording differs by band

Idea. For four dots on a line, exactly four of the seven splits work: either middle dot alone, and the two pair splits that interleave or nest; an end dot alone, or the left pair against the right pair, fails.

A shared segment, but never a shared filled triangle

Grades 2–3 P4 · first route

Idea. Two pairs on one line can share a whole segment; but a split of four labels is 2 + 2 or 1 + 3, so one group is at most a segment and the groups can never share a filled triangle.

Two arrangements with no common working split

Grades 2–3 P5 · first route

Idea. In convex position only the diagonal split works (for example AD | BC); with D inside triangle ABC only ABC | D works, so these two arrangements share no working split.

Three dots with no meeting split

Grades 2–3 P6 · return route

Idea. Three dots not on a line have no meeting split: a split is one dot against a segment, and the dot is not on that segment.

Which crosses for C give a meeting split?

Grades 2–3 P7 · return route

Idea. With A and B fixed, a split of three dots works exactly when C is on the line through A and B; the crosses on that line work and the others fail.

Two labels at one point

Grades 4–5 P4 · return route

Idea. With A and D at one point, four splits work (A | BCD, AB | CD, AC | BD, ABC | D); a split always exists, since the shared point alone against the rest meets it.

Arrangements with one, and more than one, working split

Grades 4–5 P5 · return route

Idea. In general position exactly one split works: the diagonal split of a convex quadrilateral, or the inside dot alone; a dot on the line through two others gives more than one.

Can every arrangement be split?

Grades 4–5 P6 · return route

Idea. Radon's theorem for four points in the plane: by cases (a point in the triangle of the others, a convex quadrilateral whose diagonals cross, or collinear or coincident points), a meeting split always exists.

Three dots with no meeting split

Grades 4–5 P7 · return route

Idea. Three dots not on a line have no meeting split: a split is one dot against a segment, and the dot is not on that segment.

The smallest number of dots that guarantees a split

Grades 4–5 P8 · return route

Idea. Three dots not on a line have no meeting split, and four always do, so four is the smallest number that guarantees one.